

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Instrumentation and Control Engineering) SEMESTER - 7 Winter 2025 (Regular)

Course :Bachelor of Technology (Instrumentation and Control Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTINPE702 - INSTRUMENTATION SYSTEM DESIGN

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following transducer characteristics determines how small a change in input can be detected?**
 - A. Dynamic range
 - B. Resolution
 - C. Repeatability
 - D. Sensitivity
- 2 Which signal conditioning is essential for a thermocouple before ADC conversion?**
 - A. Voltage divider
 - B. Cold junction compensation
 - C. RMS-to-DC conversion
 - D. Hysteresis correction
- 3 Which amplifier is commonly used for small RTD voltage signals?**
 - A. Instrumentation amplifier
 - B. Operational trans-conductance amplifier
 - C. Differential comparator
 - D. Schmitt trigger
- 4 A diaphragm pressure sensor converts pressure into _____**
 - A. Temperature
 - B. Angular displacement
 - C. Linear displacement
 - D. Capacitance only
- 5 An I/P converter uses which component to generate output pressure?**
 - A. Triac
 - B. Piezoelectric crystal
 - C. Flapper-nozzle system
 - D. Relay coil
- 6 MTBF is generally used for _____**
 - A. Repairable systems
 - B. Non-repairable systems
 - C. Disposable products
 - D. Short-life consumer goods
- 7 Weibull distribution is best suited for modeling**
 - A. Constant failure rate only
 - B. Wear-out failures only
 - C. Different failure behaviors including early & wear-out
 - D. Mechanical failures only

- 8 **Coaxial cables are primarily used because they offer**
 - A. Zero loss transmission
 - B. High immunity to external noise
 - C. Simple mechanical structure
 - D. Low manufacturing cost
- 9 **The best grounding method for low-noise analog circuits is**
 - A. Random grounding
 - B. Ground loops
 - C. Single-point (star) ground
 - D. No ground
- 10 **The purpose of cable shielding is to**
 - A. Reduce cable weight
 - B. Prevent EMI from entering or leaving cable
 - C. Increase voltage rating
 - D. Store static charge
- 11 **In analog circuit PCB design, sensitive signals should be routed**
 - A. Parallel to high-current lines
 - B. Close to switching regulators
 - C. Away from noise-producing components
 - D. Over split ground planes
- 12 **SMD components are preferred because they**
 - A. Increase weight
 - B. Require larger PCB area
 - C. Enable compact and automated assembly
 - D. Are only used in high-power circuits

Q2. Solve the following.

- A) Describe in detail the various testing methods used to test a transducer with calibration. 6
- B) Explain the construction, working principle and signal conditioning requirements of an LVDT-based displacement measurement system. 6

Q3. Solve the following.

- A) Explain orifice plate flow measurement system with its design steps. 6
- B) Describe the design and operation of a diaphragm-based pressure gauge. 6

Q4. Solve Any Two of the following.

- A) Explain the terms and Derive the mathematical expression for MTBF, MTTF, MTTR, Availability. 6
- B) Derive the expression for MTTF using the failure density function $f(t)$. 6
- C) Explain in details Exponential, Weibull, and Gamma reliability distributions. 6

Q5. Solve Any Two of the following.

- A) Explain the guidelines for designing enclosures used in industrial electronic systems. 6
- B) Discuss different types of cables connectors and Knobs used in instrumentation and control systems. 6
- C) Explain ESD and its harmful effects on electronic components. 6

Q6. Solve Any Two of the following.

- A) Explain in details the general guidelines for PCB component layout. 6
- B) Explain PCB Design Rules for Analog and Digital Circuits. 6
- C) Describe the soldering techniques and soldering procedure step by step. 6

*** End ***